Department of Information Science & Engineering

A Report

On

"Global Cyber Threat Landscape: Analyzing Cyberattacks and Security Incidents Worldwide"

Data Visualization Lab (BISL504)

Submitted By

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ABSTRACT

Cyberattacks have become one of the critical challenges in this digital era. While the use of the internet is engaged on a highly increasing path, organizations and governments are under enormous threat from malicious actors who wish to compromise sensitive systems. The “Global Cyberattack Analysis Dashboard” project will provide a data-driven visualization and analysis on Python’s Streamlit. This dashboard takes the pre-cleaned global cyberattacks data as input, performs aggregated analysis on frequency, severity, and target industries, and displays it in an interactive chart. Users can filter the results by country, year of the attack, type of attack, and the severity to see patterns and trends effectively.

This system utilizes modern data science tools-Pandas, Seaborn, Matplotlib, and NumPy-to manage and visualize large datasets in real time. Situational awareness for decision-makers and researchers is enhanced by the classification of attacks based on their financial and user impact into Low, Medium, and High severity levels through the dashboard. This project shows how open-source analytics tools can empower cybersecurity monitoring and education. The resulting dashboard is lightweight, accessible, and insightful; it might form one basis for subsequent machine-learning-driven threat prediction and real-time monitoring systems.

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CHAPTER 1 – INTRODUCTION

In today's digital age, cybersecurity has become essential due to the growing frequency and sophistication of online threats. This project, titled "Cyberattack Analysis Dashboard," aims to provide an interactive platform for visualizing and analyzing cyberattack data using open-source technologies such as Python, Streamlit, Pandas, Seaborn, and Matplotlib. The dashboard converts raw data into meaningful insights, enabling users to observe attack trends, types, and patterns through dynamic visualizations. It serves as an accessible and cost-effective tool for students, researchers, and professionals to better understand cybersecurity threats and make informed decisions toward preventive strategies.

1.1 Scope of work

The scope of this project will involve the design and development of a data visualization dashboard that will show a holistic view of cyberattacks around the world. The system will be targeted at analyzing historical data and not monitoring in real time. The system aims to uncover hidden insights from records of cyber incidents-enabling users to study trends across years, locations, and industries. The key deliverables expected from this project are dataset integration, severity classification, visualization using Streamlit, and creating an interactive environment for end users.

The various aspects of cybersecurity analytics the work covers include data preprocessing, filtering based on parameters, and graphical representation. This project serves as a practical example to show how data science is able to support cybersecurity research without requiring expensive infrastructure. It targets those students of cybersecurity, analysts, and policy researchers who need some easy tool to be able to study global threat patterns. This project does not try to prevent or mitigate an attack but focuses on visualization and the enhancement of awareness. Further development may extend to real-time feeds, AI-based prediction, and geospatial mapping.

1.2 Objectives

- To develop an interactive dashboard for visualizing and analyzing global cyberattacks.
- To classify cyber incidents into low, medium, and high severity levels based on impact.
- To enable users to filter and explore data by year, country, attack type, and industry.
- To identify the trend and patterns in cyber threats over time.
- To present attack insights through various visualization formats, such as charts and heat maps.

CHAPTER 2- LITERATURE SURVEY

S. No.	Publication & Year	Title / Reference	Technology Used	Features & Advantages	Limitations
1	IEEE, 2022	A Data-Driven Framework for Cyberattack Visualization and Analysis	Python, Pandas, Matplotlib	Provides interactive visualizations of global cyber incidents using real datasets.	Limited interactivity and scalability for large data volumes.
2	Springer, 2021	Machine Learning Techniques for Cyber Threat Detection	Python, Scikit-learn, TensorFlow	Uses ML algorithms to detect and classify cyber threats automatically.	Focuses mainly on prediction; lacks visualization for interpretation.
3	Elsevier, 2020	Global Cybersecurity Trends and Risk Assessment	R, Tableau	Presents statistical trends in cybercrime across industries and nations.	Static visuals; requires manual data updates.
4	ACM Digital Library, 2023	Interactive Dashboards for Cyber Threat Monitoring	Streamlit, Plotly, SQL	Enables real-time analysis and filtering of cyber incidents.	Limited integration with external APIs and datasets.
5	IEEE Access, 2022	Visualization Approaches for Cyber Threat Intelligence	Python, D3.js, Seaborn	Uses advanced data visualization to support threat analysis and decision-making.	Requires high computational resources for large datasets.

Table 2.1: Summary of literature survey

CHAPTER 3- SYSTEM ANALYSIS

3.1 Introduction

System analysis involves the identification of functional requirements, data flow, and objectives of the system. The proposed Cyberattack Dashboard will be centered on the processing of raw data from cyber incidents into useful insights. In the analysis step, it outlines what the user wants to do, like filtering, comparison, and visualization of attacks by region and type, and how these requirements can be effectively executed using Python and Streamlit.

3.2 Problem Definition

The current cybersecurity ecosystem generates unprecedented volumes of data. With so many records, unless the necessary tools are available and accessible, the stakeholder will face problems in analysis. The evolution of an attack across time, industry sectors, and countries demands visual analytics that intuitively communicate complex data. The few existing solutions in visualization are often expensive to access or require specialized technical knowledge. In general, there is also a great demand for an open-source, web-based, interactive dashboard for global cyber threat visualization.

3.3 Proposed Methodology

It proposes a methodology using Python's ecosystem for data analysis and visualization:

- Data Loading: The cleaned dataset will be read using Pandas.
- Severity Classification: A custom function that classifies the attacks into Low, Medium, and High based on financial loss and user impact.
- Filtering: Streamlit sidebar components allow users to filter the attacks by year, country, industry, and attack type.
- Visualization: This includes dynamic generation of plots such as line graphs, bar plots, pie charts, and heatmaps.
- User Interface: The Streamlit front end provides real-time updates as filters are applied.

This methodology ensures modularity, scalability, and interactivity for effective analysis.

3.4 System Requirements

Software Requirements

Operating System: Windows

Programming Language: Python 3.10 or above

Framework: Streamlit

Libraries: Pandas, Seaborn, Matplotlib, NumPy

Editor: VS Code

Hardware Requirements

Processor: Intel i3 or higher

RAM: Minimum 4 GB

Storage: 500 MB for dataset and dependencies

The system is lightweight and requires just a browser to interact with the Streamlit interface.

CHAPTER 4: SYSTEM DESIGN

4.1 Design Methodology

System design refers to the activity of using the given project reuirements to create a solution architecture that shows the system components, data flow, and interactions in an organised way. The main aim of the design phase is to organise the whole system logically and technically feasible before the actual implementation.

Design methodology in the Global Cyberattack Analysis Dashboard is aligned with the data-driven and modular principles. To ensure the clarity, maintainability, and scalability of the project, it has been segregated into various functional modules.The modlues are data collection and preprocessing, data classification, visualization, and user interaction through the Streamlit interface.

Key Design Steps:

Requirement Analysis:

The involvement of the first step is the identification of the project goals to analyse and visualise global cyberattack trends.This comprises specifying input data formts,user requirements,and visualisation goals.

Data Acquisition and Preprocessing:

The cyberattack data should be both clean and structured ,a nd are to be retrieved from the opensource repositories or datasets.Missing values,duplicates,and inconsistent entries are addressed with Python's data--processing libraries such as Pandas and NumPy.

Severity Classification:

Each deviance is assigned in one of Low,Medium,or High categories in terms of severity,dependingon the standards such as the financial loss and user impact.Conseuently,this categorisation serves as a solid foundation for the visualisation layer to be enhanced further.

Dashboard Design and Visualisation:

Streamlit is the tool of choice in creating the visual interface,which presents an interactive envirsonment where users are allowed to set filters like year,country,attack type,and severity level.In order to deict statistical relationshios clearly,graphs and charts are created with the help of Matplotlib and Seaborn.

Testing and Validation:

Usability,data accuracy,and responsiveness are the aspects that have been subjected to dashboard tests.Different scenarios have been used as test cases to verify the condition of functionality for the implemented filters,charts and data summaries.

Deployment:

After all the tests and inspections are done, the system will be ready for launch either in the local area or in the web server. Because of its simplicity in terms of structural components, the system allows users to work with it without the need for powerful computing resources.

The modular design approach structure of this project fosters the utilities of reusability, openness to change, and scalability that, in turn, make it possible to layer in future features such as ML integration and real-time monitering seamlessly.

4.2 Design Flowchart

The system flowchart shows the decision-making process that the system undergoes after receiveing the data input and until it produces te data visualisation.the flowchart delineates the flow of data through the different stages of the dashboard, thus making the users aware of the system's internal operation.

Description of Flow:

Start:

The system work is initiatedby loaded Streamit user interface.

Load Dataset:

The precleaned cyberattack dataset is loaded with the help of Pandas.

Data Preprocessing:

The data undergoes the process of filtering,missing data handling, and conversion to a new format.

Severity Classification:

Each instance of the network attack is scrutinized and is eventually classification fields having a Low, Medium, high severity of the impact.

User Input(Filter selection):

The user is able to select parameters like year, country, attack type, or severity, to limit the area of visualisation.

Data Filtering:

The system is able to filter the dataset in realtime depeding on the user-selected conditions.

Visualisation Module:

The filtered data are presented through the user of bar charts,pie cahrts,and line graphs that are generated by Seaborn and Matplotlib.

Display dahboard:

The final dashboard with the Streamlit web interface shows the realtime interactive cahrts and summaries.

End:

The user can either obtain the output or reset the filter to continue analysing data from a different point of view.

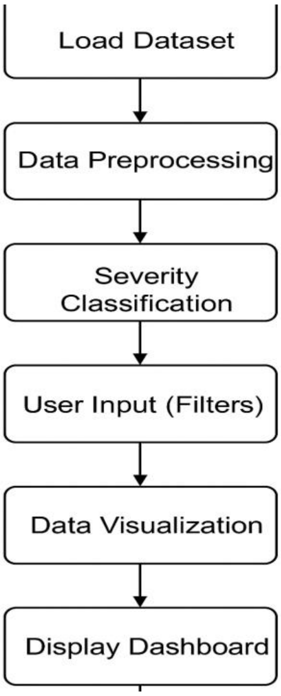

Figure 4.1: Flow Chart

CHAPTER 5: SYSTEM IMPLEMENTATION

5.1 Introduction

The implementation stage focuses on integrating the data handling, logic, and visualization components into a cohesive Streamlit app. Streamlit was chosen for its simplicity in turning Python scripts into interactive web applications without requiring any backend frameworks. The project uses a combination of Pandas for data manipulation, Matplotlib and Seaborn for plotting, with custom CSS to enhance the UI aesthetics.

5.2 Implementation of Proposed Methodology

1.Dataset Handling: Import the CSV file and preview it using st.dataframe()

2. Severity Function: This Python function evaluates each row for financial loss and affected users via conditional checks.

3. Filter Sidebar: Streamlit widgets - st.sidebar.multiselect() - provide selection options for years, severity, countries, and industries.

4.Visualizations:

   a. Line chart: Trends of attacks over years.

   b. Bar chart: Top countries affected.

   c. Pie chart: distribution of attack types.

   d. Heatmap: Attack intensity across countries and attack types.

   e. 3D-style bar chart: Industry impact.

5. Interactivity: All visuals dynamically update upon filter changes. 6. User Experience: Warnings show up when filters produce no data to ensure smooth usability.

CHAPTER 6: RESULTS

6.1 Code

```
import streamlit as st
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
import numpy as np

# Loading Cleaned Dataset

df = pd.read_csv("cyber_attacks_cleaned.csv")

# Defining Severity assignment

def assign_severity(row):
    loss = row['Financial Loss (in Million $)']

    users = row['Number of Affected Users']

    if loss > 50 and users > 100000:
        return 'High'
    elif loss > 10 and users > 10000:
        return 'Medium'
    else:
        return 'Low'

df['Severity'] = df.apply(assign_severity, axis=1)


st.title("🌍 Global Cyberattack Analysis Dashboard")
st.markdown("Interactive dashboard to explore cyber threats and security incidents worldwide.")

# Dataset Preview

st.subheader("📊 Dataset Preview")
st.dataframe(df.head(10))


# Sidebar Filters

st.sidebar.header("🔍 Filters")

years = sorted(df['Year'].unique())
selected_years = st.sidebar.multiselect("Select Year(s):", years, default=years)
```

```
if not country_attack_matrix.empty:
        fig, ax = plt.subplots(figsize=(10,6))
        sns.heatmap(
            country_attack_matrix,
            annot=True,
            cmap="YlOrRd",
            linewidths=0.5,
            linecolor='gray'
        )
        ax.set_title("Heatmap: Top 10 Countries vs Attack Types", fontsize=14)
        ax.set_xlabel("Attack Type")
        ax.set_ylabel("Country")
        st.pyplot(fig)
    else:
        st.warning("⚠ No data available to generate the heatmap.")

# Pie Chart - Attack Type Distribution

st.subheader("□ Attack Type Distribution (Pie Chart)")

if not filtered_df.empty:

    attack_type_counts = filtered_df['Attack Type'].value_counts()
    fig, ax = plt.subplots(figsize=(6,6))
    ax.pie(attack_type_counts, labels=attack_type_counts.index, autopct='%1.1f%%',
           startangle=90, colors=sns.color_palette("Set2"))
    ax.set_title("Distribution of Attack Types")
    st.pyplot(fig)

#  Cyberattacks by Affected Industry

if 'Target Industry' in df.columns:
    st.subheader("🏭 Affected Industry by Cyberattacks")

    if not filtered_df.empty:
        top_industries = filtered_df['Target Industry'].value_counts().head(10)

        fig, ax = plt.subplots(figsize=(10,5))
        bars = ax.bar(top_industries.index, top_industries.values,
                      color=plt.cm.magma(np.linspace(0.3, 0.9, len(top_industries))),
                      edgecolor='black', linewidth=1.2)

        for bar in bars:
            ax.bar(
                [bar.get_x() + bar.get_width()/2],
                [bar.get_height()*0.05],
                bottom=-bar.get_height()*0.05,
                width=bar.get_width()*0.9,
```

```
    color='gray',
                alpha=0.3
            )
            bar.set_edgecolor('black')
            bar.set_linewidth(1.3)

        plt.xticks(rotation=45, ha='right')
        ax.set_title("Top 10 Affected Industries", fontsize=14, fontweight='bold')
        ax.set_ylabel("Number of Attacks")

        for i, v in enumerate(top_industries.values):
            ax.text(i, v + max(top_industries.values)*0.01, str(v),
                    ha='center', va='bottom', fontweight='bold', color='black')

        ax.grid(axis='y', linestyle='--', alpha=0.4)
        st.pyplot(fig)
    else:
        st.info("⚠ No data for the selected filter combination.")
else:
    st.info("⚠ 'Target Industry' column not found in the dataset.")
```

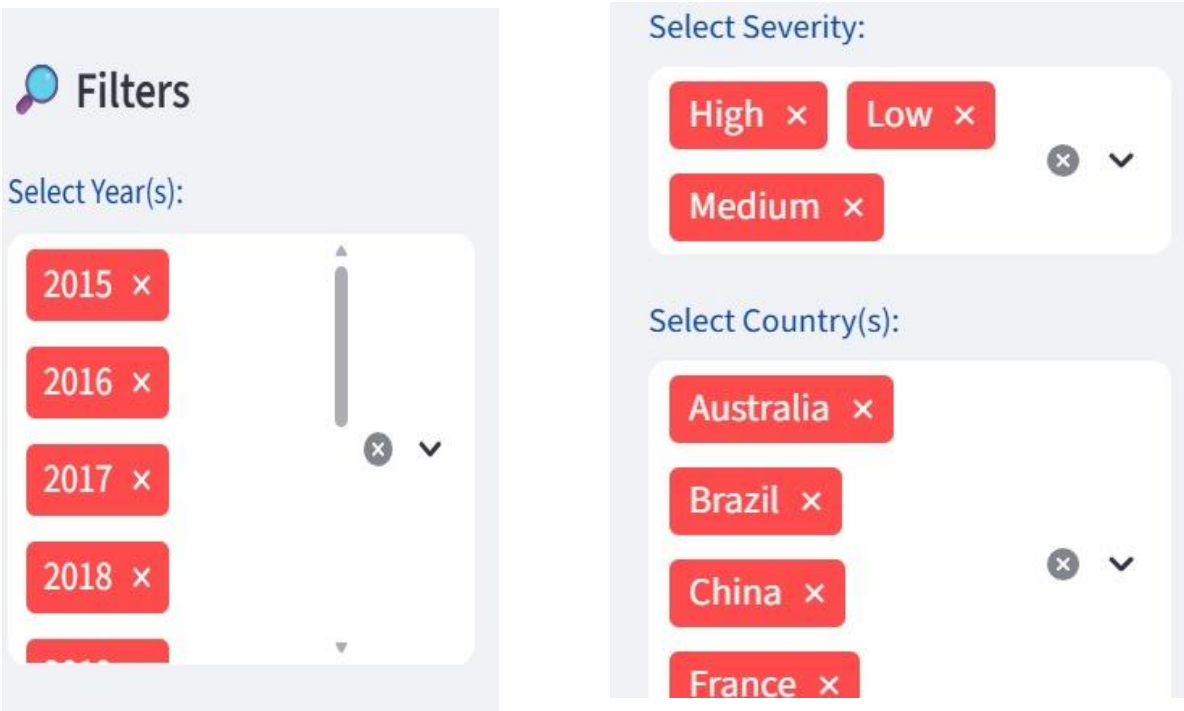

Figure 6.1: Filters

	Country	Year	Attack Type	Target Industry	Financial Loss (in Million $)	Number of Affe
0	China	2019	Phishing	Education	80.53	
1	China	2019	Ransomware	Retail	62.19	
2	India	2017	Man-in-the-Middle	IT	38.65	
3	UK	2024	Ransomware	Telecommunications	41.44	
4	Germany	2018	Man-in-the-Middle	IT	74.41	
5	Germany	2017	Man-in-the-Middle	Retail	98.24	
6	Germany	2016	DDoS	Telecommunications	33.26	

Figure 6.2: Dataset preview

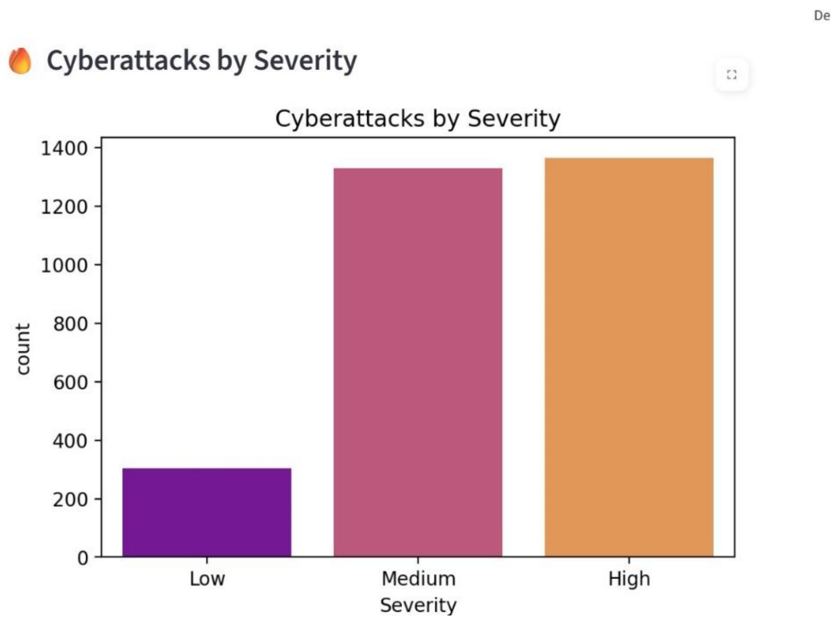

Figure 6.3: Bar chart for cyberattacks by severity

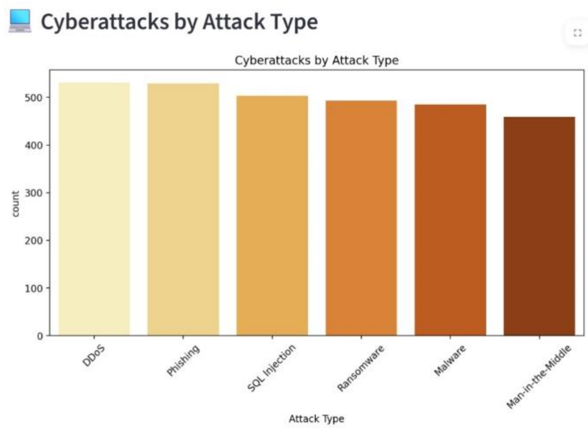

Figure 6.4: Bar Chart for cyberattacks by Attack type

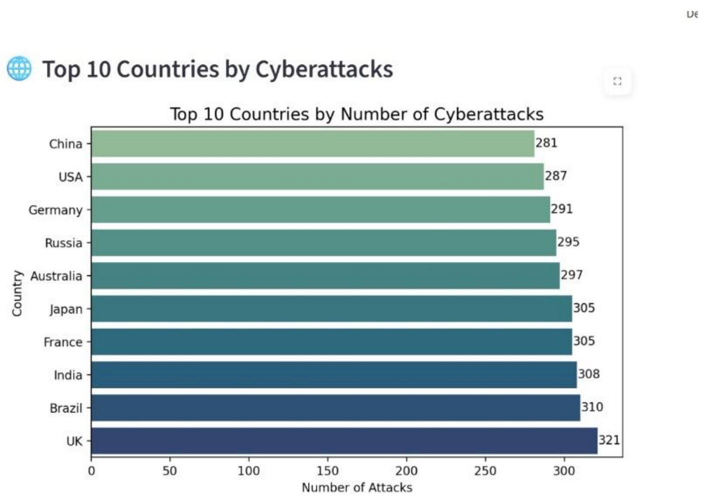

Figure 6.5: Bar chart for top 10 countries by cyberattacks

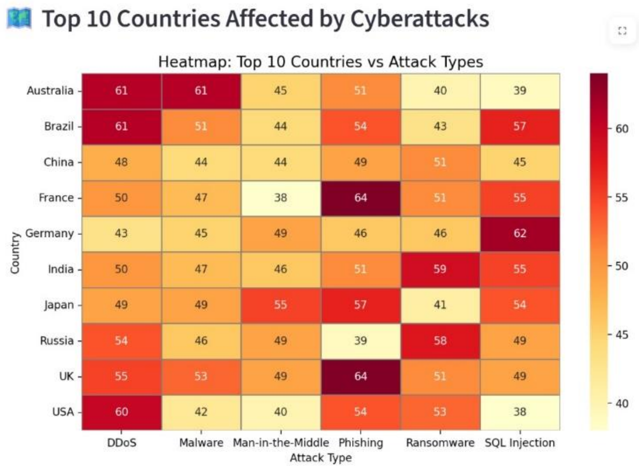

Figure 6.6: Heatmap for top 10 countries vs attack types

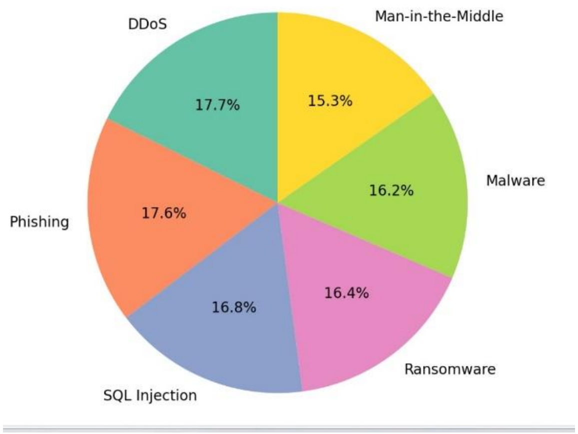

Figure 6.7: Pie chart for Attack Types

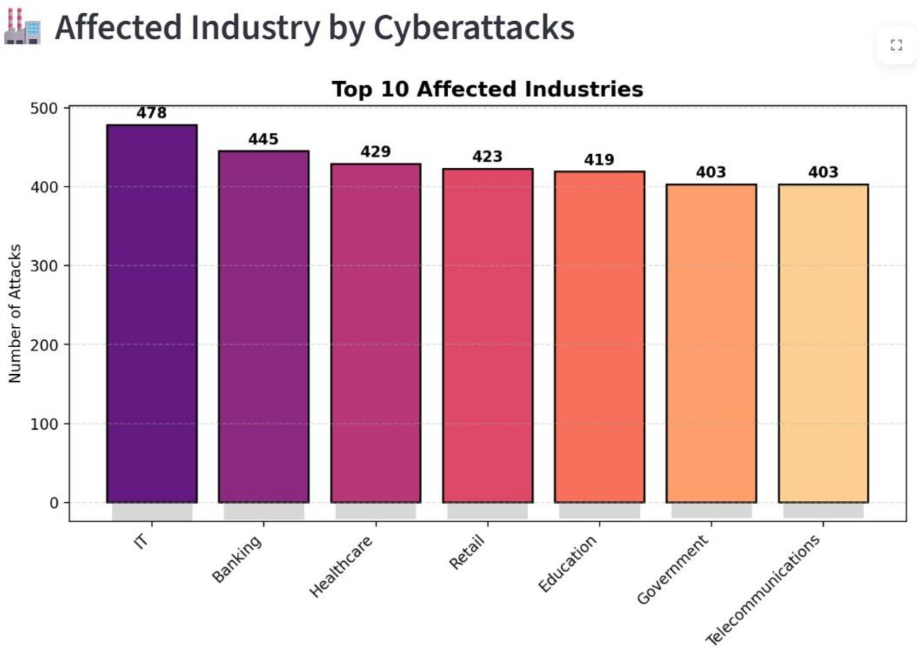

Figure 6.8: Bar chart for Affected industries

CONCLUSION

The "Global Cyber Threat Landscape: Analyzing Cyberattacks and Security Incidents Worldwide using Streamlit" project indeed illustrates the use of Python-based visualization to analyze and interpret global cybersecurity data. It provides an interactive data-driven system that enables users to understand the patterns of cyberattacks by year, country, industry, and severity. Thus, it converts static datasets into dynamic visuals, simplifying complex cybersecurity data and allowing deeper insight generation for effective analysis and awareness.

This project has the major advantage of being highly cost-effective and accessible, since it uses only open-source technologies like Streamlit, Pandas, Seaborn, and Matplotlib. These will make possible a very visually rich analytical dashboard with no costly enterprise software use. The platform is user-friendly, scalable, and adaptable to future enhancements; as such, it suits students, researchers, and professionals alike. However, it has the minor limitation of using static datasets, limiting real-time threat monitoring. Besides, there is a slight performance difference on very large datasets or when using complex visualizations.

Overall, the project meets its core objective by offering an interactive, affordable, and insightful way to analyze cybersecurity data. By bringing simplicity, effectiveness, and flexibility, it allows users to identify major trends of threats and gain insightful insights into the evolving global cyber landscape in a way that is truly contributing to higher digital security awareness and preparedness.

FUTURE ENHANCEMENTS

- Integration of real-time data feeds using APIs.
- Machine learning to implement predictive analytics.
- Addition of geospatial visualizations using Plotly or Folium.
- Exportable reports and automated insight summaries.
- Cloud deployment for multi-user access.

The project thus sets a solid basis for further development into a holistic cyber threat intelligence system.

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